

Unload a Dish Rack into a Cabinet Organizer SOP

One episode empties one loaded dish rack into one cabinet organizer. Nine washed items sit in the rack, still damp from the wash: a large plate, a small plate, two bowls, two cups with handles, a fork, a knife, and a spoon. Every one of them is dried on a towel before it is put away, and every one of them has a fixed place in the organizer.

The rack is loaded differently every episode. Which slot each plate stands in, where the bowls and cups sit, and whether something is resting over something else all change from take to take. So **the unload order is fixed by item type, not by position**: plates, then bowls, then cutlery, then cups. That one rule is what keeps the data consistent across a scene that is never the same twice.

Every item comes out of the rack in the right gripper, is wiped twice with the towel, and is **handed across to the left gripper in the air** before the left gripper lowers it into its place in the organizer. Where the wipe happens depends on the item: **plates and bowls are set on the drying mat in the middle of the table and wiped there**, then lifted back off it and handed over, while **cups and cutlery are wiped in the air** and never touch the mat at all. No item skips the wipe, and no item reaches the organizer without going through the handoff.

The organizer has two shelves. **Plates and bowls go on the upper shelf, the cutlery goes on the lower shelf, and the cups hang from hooks on the front edge of the upper shelf.**

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the cabinet organizer on the left, the drying mat in the middle, the towel in its home spot, and the loaded dish rack on the right.
3. Both shelves of the organizer and the cup hooks are in view, and the space in the air over the drying mat is in view, so the handoff and the in air wipes can be seen.
4. Both arms are at home with grippers open.

5. The tabletop is dry outside the mat and the rack.
6. Nothing obstructs the paths between the rack, the drying mat, and the organizer.

Materials checklist

1. The dish rack sits against the right edge, loaded with all nine items: a large plate, a small plate, two bowls, two cups with handles, a fork, a knife, and a spoon.
2. Plates stand upright in the rack slots. Bowls and cups sit inverted on the rack shelf. The three utensils stand handle up in the rack basket.
3. Every item is damp. No item is dripping and there is no standing water in the rack tray.
4. The load is uneven, as a real wash leaves it: items at no fixed spacing, and at least one bowl or cup sitting partly over a plate.
5. The wiping towel is dry, folded into a flat pad, and lying in its home spot to the right of the drying mat.
6. The drying mat lies flat in the middle of the tabletop and is empty.
7. The cabinet organizer is empty: nothing anywhere on the upper shelf, the cup hooks bare, and all three cutlery compartments bare.
8. The gap between the upper and lower shelf is clear, so a cup can hang from a hook without touching the lower shelf.

Workspace layout

- **Cabinet organizer:** on the tabletop against the left edge, within reach of the left arm. It has two shelves, one above the other:
 - **Upper shelf:** one flat open shelf, undivided, that items rest on directly. It holds the plate stack and the nested bowls side by side, clear of each other.
 - **Cup hooks:** hooks fixed to the front edge of the upper shelf. A cup hung on a hook hangs down into the open gap between the two shelves.
 - **Lower shelf:** holds the cutlery tray with its three compartments in a row.
- **Drying mat:** flat in the middle of the tabletop, reachable by both arms. It holds exactly one plate or bowl at a time and nothing else, and is empty at the end of the episode. Cups and cutlery are wiped in the air and never rest on it.
- **Handoff point:** in the air directly over the drying mat, about a hand's height above it. Every item changes arms here, and cups and cutlery are also wiped here. Anything that slips at the handoff or during an in air wipe falls onto the mat rather than onto the bare tabletop.
- **Towel home:** the clear patch of tabletop just right of the drying mat, within reach of the right arm. The towel lies here whenever it is not being used.
- **Dish rack:** on the tabletop against the right edge, within reach of the right arm.

Each arm owns the side it sits on. The right arm works the rack, the towel, and the mat, the left arm works the organizer, and every item changes arms at the handoff point over the mat.

Vocabulary

Arm assignments

- **The right gripper owns the rack, the towel, and the mat.** It lifts every item out of the rack, sets the plates and bowls on the drying mat, does every wipe, lifts each wiped plate and bowl back off the mat, and hands every item to the left gripper. It never reaches the organizer.
- **The left gripper owns the organizer.** It holds each item still for its wipe, on the mat for a plate or bowl and in the air for a cup or utensil, takes the item from the right gripper at the handoff, and lowers it into its place in the organizer. It never reaches the rack, never touches the towel, and never lifts an item off the mat itself.
- **Every item is passed hand to hand.** The right gripper is the only arm that lifts an item off the mat, and the left gripper only ever receives an item in the air at the handoff point.
- An item is released only into the left gripper at the handoff, onto the drying mat, or into its place in the organizer. Never open a gripper over open space, and never over the rack once an item has left it.

The drying cycle

Every one of the nine items is wiped twice with the towel and handed from the right gripper to the left gripper before it is put away. **Where the wipe happens depends on the item type.**

- **Plates and bowls are wiped on the mat.** They are set down on the drying mat, wiped there, then lifted back off it and handed over.
- **Cups and cutlery are wiped in the air.** They come straight out of the rack to the handoff point, are handed over there, and are wiped while the left gripper holds them up. They never touch the mat.

In both cycles the left gripper is the arm that holds the item still for the wipe, the right gripper is the only arm that touches the towel, and every wipe is exactly two passes. Two passes is the standard, not a dry surface. Whatever moisture is left after two passes stays. Both cycles wipe over the mat, so drips and anything that slips land on the mat rather than on the bare tabletop.

Cycle A, plates and bowls: place, wipe, hand over, put away.

1. **Place it.** The right gripper lifts the item straight up out of the rack, carries it level to the drying mat, and lowers it into the drying pose for its type. It releases once the item is resting still.
2. **Wipe it on the mat.** The left gripper takes hold of the item and holds it still against the mat. The right gripper pinches the towel in its home spot, lifts it, presses it onto the item, and

draws it across the item **twice**. It then lays the towel back flat in its home spot and releases. The left gripper keeps its hold the whole time.

3. **Hand it over.** The left gripper releases the item and comes clear of it. The right gripper closes on the item, lifts it straight up off the mat to the **handoff point**, and holds it still there. The left gripper then closes on it in the grip it will carry and place it by. Only once the left gripper is closed and holding does the right gripper open and come away.
4. **Put it away.** The left gripper carries the item level from the handoff point into the organizer, lowers it into its place on the **upper shelf**, and releases.

Cycle B, cups and cutlery: hand over, wipe, put away.

1. **Lift it and hand it over.** The right gripper lifts the item straight up out of the rack and carries it level to the **handoff point** in the pose its step calls for, without touching the mat on the way, and holds it still there. The left gripper then closes on it in the grip it will carry and place it by. Only once the left gripper is closed and holding does the right gripper open and come away.
2. **Wipe it in the air.** The left gripper holds the item still at the handoff point, clear of the mat. The right gripper pinches the towel in its home spot, lifts it, presses it onto the item, and draws it across the item **twice**. It then lays the towel back flat in its home spot and releases. The item is not set down, moved, or turned at any point during the wipe.
3. **Put it away.** The left gripper carries the item level from the handoff point to its place, hangs the cup on its hook or lowers the utensil into its compartment on the **lower shelf**, and releases.

The handoff is the same exchange in both cycles. The item is stationary at the handoff point for the whole of it. Both grippers are closed on the item at the same time for a moment, the right one opening only after the left one has it. Neither arm pulls, twists, or lifts the item while the other is still closed on it, and the item is never in free air with no gripper holding it.

One carrying grip per item. The grip the left gripper takes at the handoff is the grip it carries and places the item by. Do not let go and take a fresh grip between the handoff and the put away, and do not set an item back down once it has been handed over.

Working a loaded rack

- **Type order, not position order.** Unload plates, then bowls, then cutlery, then cups, no matter where each item happens to be sitting. Do not unload by whatever is nearest or easiest.
- Within the plates, the order is always **large plate, then small plate**. Within the bowls and the cups, take the item **nearer the front of the rack first**. Cutlery is the one exception to front first: it always goes **fork, knife, spoon**.
- **Never pull an item out from under another.** If the item due next is covered or blocked, lift the covering item straight up first, set it clear on the rack shelf, and then take the item that

was due. This is the only permitted departure from the order, and the order resumes from the top of the type you were on.

- Take each grip where the item is clear, not where the standard grip would be on an empty rack.
- Never drag an item along a rack slot or across the rack shelf to get a better grip. Lift it straight up.

Organizer places

- **Plates: upper shelf.** Stacked flat and face up directly on the shelf, large plate on the bottom, small plate centered on top of it. Where on the shelf the stack goes is not prescribed.
- **Bowls: upper shelf.** Nested face up directly on the shelf beside the plate stack, the first bowl down flat and the second bowl lowered inside it. The bowls touch neither the plate stack nor a shelf edge; which side of the stack they go is not prescribed.
- **Cups: hung from the front edge of the upper shelf.** Hanging with the handle over a hook, the body hanging free in the gap between the two shelves and touching neither the lower shelf nor the other cup. Each cup goes on a free hook, one cup per hook.
- **Cutlery: lower shelf.** Lying flat in the cutlery tray, one utensil per compartment, each utensil wholly inside its own compartment and clear of the dividers. Which compartment a given utensil goes in, and which way it points along that compartment, are not prescribed.

Nothing rests on the organizer outside these places, nothing is set on top of the upper shelf, nothing but the cutlery goes on the lower shelf, and nothing overhangs an edge.

Handling standard

- Move exactly one item at a time.
- The towel is never carried in the same gripper as an item, and never in the same trip.
- Carry every item level and lower it straight down.
- Lower each item until it is resting on the mat, on the shelf, in its tray compartment, or on the item below it, then release. Never drop an item from above.
- Nothing is slid. The only sliding motion in the whole episode is the towel drawn across an item during a wipe. Correct a bad placement by lifting the item and setting it down again.
- An item is settled when it stays still for 2 seconds after release. A hung cup is settled when it has stopped swinging.

Steps

Step 1: Dry and stack the two plates

Goal: both plates are dry and stacked face up on the upper shelf.

Plates run **Cycle A**: wiped on the mat, then handed over. Work one plate at a time in this order, wherever each one is standing: **large plate, then small plate**.

1.1 Place it on the mat

- If a bowl or cup is resting over the plate due next, lift that item straight up and set it clear on the rack shelf first.
- The **right gripper** grips the plate by any free stretch of rim, lifts it straight up out of the slot, and carries it level to the drying mat.
- Lower it flat and face up in the middle of the mat, then release.

1.2 Wipe it on the mat

- The **left gripper** takes the plate by its rim and holds it still against the mat.
- The **right gripper** lifts the towel from its home spot, presses it onto the face of the plate, and draws it across the face twice.
- The **right gripper** lays the towel back flat in its home spot and releases.

1.3 Hand it over

- The **left gripper** releases the plate and comes clear of it.
- The **right gripper** grips the plate by its rim, lifts it straight up off the mat to the **handoff point**, and holds it level and still.
- The **left gripper** closes on the far side of the plate rim, then holds it while carrying it by, keeping the plate level.
- The **right gripper** opens and comes away only once the left gripper is holding the plate.

1.4 Stack it on the upper shelf

- The **left gripper** carries the plate level from the handoff point to the **upper shelf**.
- Lower it face up. The large plate goes flat on the shelf, clear of the shelf edges and with room beside it for the bowls; the small plate is centered on top of the large one.
- Release once the plate is resting.

Check: both plates are face up on the upper shelf, large on the bottom, small centered on top, and the stack sits square rather than leaning. If it leans or the top plate rocks, lift that plate and lower it again. Do not shove it into place.

Step 2: Dry and nest the two bowls

Goal: both bowls are dry and nested face up on the upper shelf beside the plate stack.

Bowls run **Cycle A**: wiped on the mat, then handed over. Work one bowl at a time, taking the **bowl nearer the front of the rack first**.

2.1 Place it on the mat

- If a cup or utensil is resting over the bowl due next, lift that item straight up and set it clear on the rack shelf first.
- The **right gripper** grips the bowl by any free stretch of rim, lifts it straight up off the rack shelf, and carries it level to the drying mat.
- Lower it face up in the middle of the mat, then release.

2.2 Wipe it on the mat

- The **left gripper** takes the bowl by its rim and holds it still against the mat.
- The **right gripper** lifts the towel from its home spot, presses it onto the outside of the bowl, and draws it across the outside twice.
- The **right gripper** lays the towel back flat in its home spot and releases.

2.3 Hand it over

- The **left gripper** releases the bowl and comes clear of it.
- The **right gripper** grips the bowl by its rim, lifts it straight up off the mat to the **handoff point**, and holds it level and still.
- The **left gripper** closes on the far side of the bowl rim, then hold it will carry it by, keeping the bowl level.
- The **right gripper** opens and comes away only once the left gripper is holding the bowl.

2.4 Nest it on the upper shelf

- The **left gripper** carries the bowl level from the handoff point to the **upper shelf**.
- Lower it face up beside the plate stack, touching neither the plates nor a shelf edge. The first bowl goes flat on the shelf; the second is lowered inside the first until it is resting on it.
- Release once the bowl is resting.

Check: both bowls are face up on the upper shelf, clear of the plate stack, and the second sits down inside the first rather than perched on its rim. If the second bowl sits high or tips, lift it and lower it again.

Expected state: the rack still holds the two cups and the three utensils. The mat is bare, the towel is flat in its home spot, and the upper shelf holds the plate stack and the nested bowls side by side. Nothing has been put on the lower shelf yet.

Step 3: Dry and stow the three utensils

Goal: the fork, knife, and spoon are dry and lying one per compartment in the cutlery tray on the lower shelf.

Cutlery runs **Cycle B**: handed over first, then wiped in the air. A utensil does not touch the drying mat at any point. Work one utensil at a time in this order, wherever each one is standing

in the basket: **fork, knife, spoon.**

- **Lift and hand over:** the **right gripper** grips the utensil across its **handle**, lifts it straight up out of the basket, and brings it to the **handoff point** over the mat, held level. The **left gripper** closes on the **shaft**, the stretch between the handle and the head, keeping the utensil level. The **right gripper** opens and comes away only once the left gripper is holding the utensil.
- **Wipe in the air:** the **left gripper** holds the utensil level and still at the handoff point, clear of the mat. The **right gripper** draws the towel along the utensil from the handle end to the tip twice, over the left gripper's hold, then returns the towel to its home spot.
- **Put away:** the **left gripper** carries the utensil level to the **lower shelf** and lowers it flat into an empty compartment of the cutlery tray. Which compartment it goes in, and which way it points along that compartment, are not prescribed.

Check: one utensil per compartment in the cutlery tray on the lower shelf, each lying flat and wholly inside its own compartment. No compartment holds two utensils, and no utensil rests on a tray divider or overhangs the tray.

Expected state: the rack still holds the two cups. The mat is bare, the towel is flat in its home spot, the upper shelf holds the plate stack and the nested bowls, and the three utensils lie in the cutlery tray on the lower shelf. The cup hooks are still bare.

Step 4: Dry and hang the two cups

Goal: both cups are dry and hanging from the hooks on the front edge of the upper shelf.

Cups run **Cycle B:** handed over first, then wiped in the air. A cup does not touch the drying mat at any point. Work one cup at a time, taking the **cup nearer the front of the rack first**. Each cup goes on a free hook, one cup per hook.

4.1 Lift it and hand it over

- The **right gripper** grips the cup around its body, lifts it straight up off the rack shelf, and carries it to the **handoff point** over the mat with the **handle pointing left**, toward the organizer. It holds the cup still there.
- The **left gripper** closes on the cup around its body, the hold it will carry and hang it by, keeping the handle pointing away from the arm.
- The **right gripper** opens and comes away only once the left gripper is holding the cup.
- The cup is not set down on the mat, the tabletop, or the organizer between the rack and the hook.

4.2 Wipe it in the air

- The **left gripper** holds the cup still at the handoff point, clear of the mat.

- The **right gripper** lifts the towel from its home spot and draws it down the side of the cup twice.
- The **right gripper** lays the towel back flat in its home spot and releases.
- The cup does not swing, turn, or dip while the towel is on it. If it turns in the left gripper's hold, hold it steady and finish the two passes rather than regripping it.

4.3 Hang it

- The **left gripper** carries the cup level from the handoff point to the **front edge of the upper shelf**, still with the handle pointing away from the arm.
- Lower the cup so the handle drops over a free hook, then release only once the cup is hanging and has stopped swinging. Keep the cup clear of the cutlery tray and the utensils already lying on the lower shelf as it goes down.
- If the handle misses the hook or the cup swings off, catch it or lift it clear and hang it again. If it has come to rest anywhere else, the right gripper picks it up and hands it over again.

Check: both cups hang still, one cup per hook, each hanging free in the gap between the shelves. Neither cup rests on the upper shelf, on the lower shelf or its cutlery tray, or against the other cup.

Expected state: the rack is empty, the mat is bare, the towel is flat in its home spot, and the organizer holds the plate stack and the nested bowls on the upper shelf, the three utensils on the lower shelf, and the two cups hanging from the front edge.

Step 5: End the episode

Goal: the rack is empty, the organizer is fully loaded, and both arms are safely home.

- Confirm the rack holds nothing, including the basket, and that no item was left on the mat or on the bare tabletop.
- Confirm the towel is lying flat in its home spot and not on the mat, in the rack, or in the organizer.
- Confirm every place in the organizer is filled as prescribed: the plate stack and the nested bowls on the upper shelf, the three utensils on the lower shelf, and both cups hanging from the front edge hooks.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unload the organizer completely: the upper shelf, the hooks on its front edge, and all three cutlery compartments on the lower shelf.

2. Wet all nine items again so they are damp but not dripping, and shake off any standing water.
3. Reload the rack: plates upright in the slots, bowls and cups inverted on the shelf, the three utensils handle up in the basket. Vary the arrangement between episodes rather than rebuilding the same load, and leave at least one bowl or cup sitting partly over a plate.
4. Empty the rack tray if water has collected in it.
5. Replace the wiping towel with a dry one, folded into a flat pad, and lay it in its home spot.
6. Wipe the tabletop and clear the drying mat.
7. Clean or replace anything chipped, cracked, bent, or stained.
8. Check the cup hooks are tight on the front edge of the upper shelf and the gap between the two shelves is clear, so a hung cup hangs free of the lower shelf.
9. Put the organizer back against the left edge and the rack back against the right edge.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Unloaded by position instead of by type

- **Visible cue:** items come out of the rack in whatever order they happen to be reachable, such as a cup before the plates, a cup before the cutlery, or the cutlery split around a bowl, rather than plates, then bowls, then cutlery, then cups.
- **SOP rule broken:** Steps 1 to 4, unload by type order: plates, bowls, cutlery, cups, no matter where each item is sitting.
- **Coaching note:** the load changes every episode; the order never does.

Violation: Wrong order within a type

- **Visible cue:** the small plate is taken before the large one, the bowl or cup at the back of the rack is taken before the one at the front, or the utensils leave in any order other than fork,

knife, spoon.

- **SOP rule broken:** Steps 1 to 4, large plate then small plate, bowls and cups front of the rack first, and cutlery always fork, knife, spoon.
- **Coaching note:** name the next item before you reach for it.

Violation: Item pulled out from under another

- **Visible cue:** a plate is pulled out from under a bowl or cup resting over it, or an item is slid sideways along a slot or across the rack shelf to free it, instead of the covering item being lifted straight up and set clear first.
- **SOP rule broken:** Step 1.1, never pull an item out from under another, lift the covering item straight up, set it clear on the rack shelf, then take the item that was due.
- **Coaching note:** uncover it, then lift it. Never tug it free.

Violation: Arm crossed into the other arm's side

- **Visible cue:** the left gripper reaches into the rack, lifts an item out of it, lifts an item off the mat, or picks up the towel; or the right gripper reaches the organizer or puts an item away.
- **SOP rule broken:** Steps 1 to 4, the right gripper owns the rack, the towel, and the mat, the left gripper owns the organizer, and items change arms at the handoff point.
- **Coaching note:** right arm right of the mat, left arm left of it. Only the right arm takes an item off the mat.

Violation: Handoff skipped

- **Visible cue:** the left gripper lifts an item off the mat itself, or the right gripper carries an item toward the organizer, instead of the item being exchanged at the handoff point over the mat.
- **SOP rule broken:** The drying cycle, every item is handed from the right gripper to the left gripper in the air at the handoff point.
- **Coaching note:** nothing reaches the organizer except through the left gripper's hand, and nothing gets into the left gripper except at the handoff.

Violation: Handoff released early or fumbled

- **Visible cue:** the right gripper opens before the left gripper is closed on the item, the item slips or falls during the exchange, both arms tug or twist the item while both are closed on it, or the item is moved through the air by one arm while the other is still holding it.
- **SOP rule broken:** The drying cycle, the item is stationary at the handoff point, both grippers hold it at once for a moment, and the right one opens only after the left one has it.
- **Coaching note:** hold still, let the other gripper close, then let go. Never hand off on the move.

Violation: Wiped in the wrong place for the item type

- **Visible cue:** a plate or bowl is wiped while held up in the air rather than resting on the mat, or a cup or utensil is set down on the mat, the tabletop, or the organizer and wiped there rather than being wiped in the air at the handoff point.
- **SOP rule broken:** The drying cycle, plates and bowls are wiped on the mat (Cycle A) and cups and cutlery are wiped in the air (Cycle B).
- **Coaching note:** flat things go down to be wiped, cups and cutlery stay up.

Violation: Item put away without being dried

- **Visible cue:** an item reaches the organizer with no towel pass at all, or a plate or bowl goes from the rack to the handoff point without being set on the mat and wiped there.
- **SOP rule broken:** Steps 1 to 4, every item is wiped twice before it is put away, plates and bowls on the mat and cups and cutlery in the air.
- **Coaching note:** no item skips the towel, whichever cycle it is on.

Violation: Wrong number of towel passes

- **Visible cue:** the towel is drawn across the item once, or three or more times, instead of exactly twice; or the wiping continues to chase a dry surface.
- **SOP rule broken:** Steps 1 to 4, draw the towel across the item twice, two passes is the standard, not a dry surface.
- **Coaching note:** count the two passes, then stop, whatever the item still looks like.

Violation: Item not held during the wipe

- **Visible cue:** the right gripper wipes an item that no gripper is holding, and the item shifts, spins, is pushed across the mat by the towel, or swings on the end of the left gripper's hold during an in air wipe.
- **SOP rule broken:** Steps 1 to 4, the left gripper takes hold of the item, holds it still, and keeps its hold for the whole wipe.
- **Coaching note:** the left hand anchors the item before the towel touches it.

Violation: Item regripped after the handoff

- **Visible cue:** the left gripper releases the item after taking it at the handoff and takes a fresh grip before or after carrying it, sets it back down on the mat or the tabletop, or hands it back to the right arm to be repositioned.
- **SOP rule broken:** The drying cycle, the grip the left gripper takes at the handoff is the grip it carries and places the item by.
- **Coaching note:** take the carrying grip at the handoff, then go straight to the place.

Violation: Towel mishandled

- **Visible cue:** the towel is left on the drying mat, dropped on the tabletop or floor, carried in the same gripper as an item, put into the rack or the organizer, or left anywhere other than flat in its home spot between wipes.
- **SOP rule broken:** Steps 1 to 4, lay the towel back flat in its home spot after every wipe, and never carry it in the same gripper as an item.
- **Coaching note:** the towel goes home after every single wipe.

Violation: Item put in the wrong place

- **Visible cue:** a plate or bowl is set on the lower shelf, a utensil put on the upper shelf, a cup set on either shelf instead of hung, two utensils sharing a compartment, a bowl set on or against the plate stack instead of clear of it, or any item left on the organizer outside its place.
- **SOP rule broken:** Steps 1 to 4 and Organizer places, plates stacked and bowls nested side by side on the upper shelf, cups hung from the front edge hooks, and one utensil per compartment on the lower shelf.
- **Coaching note:** plates and bowls up, cutlery down, cups hanging. Say the place before you carry the item.

Violation: Plate stack built incorrectly

- **Visible cue:** the small plate is on the bottom, a plate is face down, the top plate is clearly off center, or the stack leans and is left that way.
- **SOP rule broken:** Step 1.4 and Organizer places (large plate flat on the bottom, small plate centered face up on top; lift and lower again if it leans or rocks).
- **Coaching note:** square the small plate on the large one before releasing.

Violation: Bowls not nested

- **Visible cue:** the second bowl is set beside the first, perched on its rim, or placed face down, instead of being lowered inside it.
- **SOP rule broken:** Step 2 and Organizer places (bowls nested face up, the second lowered inside the first until it rests on it).
- **Coaching note:** take the second bowl all the way down inside the first.

Violation: Cup hung incorrectly

- **Visible cue:** a cup is left standing on either shelf, hung on a hook that already holds the other cup, left resting on the lower shelf or against the other cup instead of hanging free, or released while it is still swinging or before the handle is over the hook.
- **SOP rule broken:** Step 4.3 and Organizer places (hang from a free hook on the front edge with the handle over the hook and the body hanging free in the gap between the shelves, one cup per hook, release only once it hangs still).

- **Coaching note:** watch the handle seat on the hook, then wait for it to stop moving.

Violation: Utensil not seated flat in its compartment

- **Visible cue:** a utensil lies across two compartments, rests on top of a tray divider, sticks up out of its compartment, or is left propped against the side rather than lying flat inside it.
- **SOP rule broken:** Step 3 and Organizer places (each utensil lies flat and wholly inside its own compartment, clear of the dividers).
- **Coaching note:** take it all the way down into the channel before you open the gripper.

Violation: More than one item moved at once

- **Visible cue:** two utensils, two bowls, or a plate with a cup still resting on it are lifted together, or an item is carried while the towel is still in the other gripper mid wipe.
- **SOP rule broken:** Steps 1 to 4, move exactly one item at a time.
- **Coaching note:** one item per trip, every trip.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the mat, a shelf, or a tray compartment and the item falls the rest of the way, instead of being lowered until it is resting.
- **SOP rule broken:** Steps 1 to 4, lower each item until it is resting, then release, and never drop an item from above.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, nudges, or drags an item along a rack slot, across the rack shelf, across the mat, or across the organizer, either to get a better grip or to correct a placement, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 1 to 4, nothing is slid except the towel during a wipe, and correct a bad placement by lifting the item and setting it down again.
- **Coaching note:** lift it clear and put it down again; nothing gets shoved into position.

Violation: Mat or rack left occupied

- **Visible cue:** the episode reaches Step 5 with an item still on the drying mat, still in the rack or its basket, or sitting on the bare tabletop.
- **SOP rule broken:** Step 5 (the rack holds nothing and the mat is bare before the episode ends).
- **Coaching note:** sweep the rack, the basket, and the mat with your eyes before homing.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a place in the organizer unfilled, the towel out of its home spot, an arm away from home.

- **SOP rule broken:** Step 5 (confirm the empty rack, the bare mat, the towel at home, and the fully loaded organizer; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective object:** a chipped, cracked, or bent item found mid episode, a cup hook that pulls loose, or a rack or organizer that shifts off its zone. Replace it before the next episode.
- **Towel saturated before the episode ends:** the towel stops picking up water even though the prescribed two passes were run. Swap in a dry towel before the next episode.

Annotation subtasks (from SOP)

1. Lift a covering item off a plate and set it clear on the rack shelf
2. Move one plate from the rack to the drying mat
3. Move one bowl from the rack to the drying mat
4. Hold a plate or bowl still on the mat during the wipe
5. Wipe one item with the towel and return the towel to its home spot
6. Lift a wiped plate or bowl off the mat and hand it to the left gripper at the handoff point
7. Move one utensil from the rack basket to the handoff point and hand it to the left gripper
8. Move one cup from the rack to the handoff point and hand it to the left gripper
9. Hold a cup or utensil still in the air during the wipe
10. Stack one plate on the upper shelf
11. Nest one bowl on the upper shelf beside the plate stack
12. Lay one utensil flat in an empty compartment of the cutlery tray on the lower shelf
13. Hang one cup on a hook on the front edge of the upper shelf
14. Return both arms home and end the episode